

Qatar-2b

R-1956

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_130413 · created 2026-08-06T16:04:02+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456395.78306 +/- 0.00026 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.206 +/- 0.058
Transit depth (Rp/Rs)^2	4.24 +/- 2.39 [%]
Orbital Inclination (inc)	84.81 +/- 1.7
Airmass coefficient 1 (a1)	0.931 +/- 0.063
Airmass coefficient 2 (a2)	0.054 +/- 0.071
Scatter in the residuals of the lightcurve fit is	6.71 %
Best Comparison Star	#1 - [507, 40]
Optimal Aperture	3.12
Optimal Annulus	12.19
Transit Duration (day)	0.0685 +/- 0.0095

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.206 $\pm$ 0.058 vs 0.16327 (deviation 0.74 σ)
Duration fit vs published	0.0685 d vs 0.0767 d (deviation 0.86 σ)
Mid-time O-C	-0.5 min
Depth SNR	3.6
Residual scatter	6.71 %

Light curve

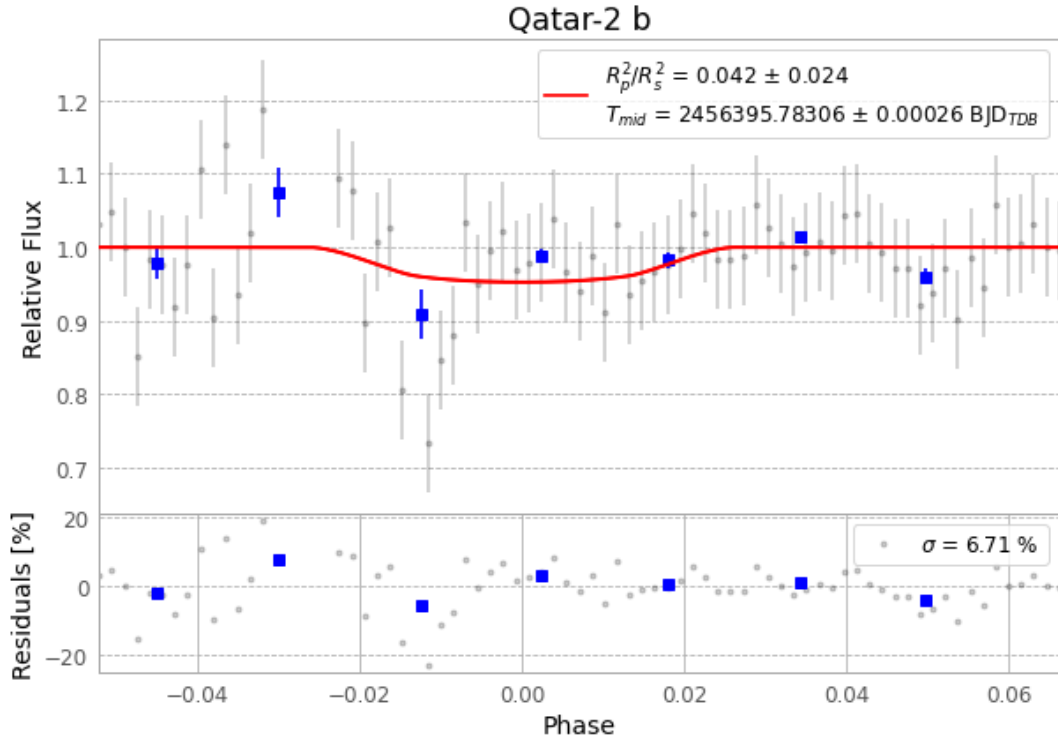


Figure 1 — FinalLightCurve_Qatar-2 b_2013-04-12.png (SHA-256 3c56c5ceb6790f2f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [259, 265], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6a124b9443a026a5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

77 FITS frames (51 MB), Qatar-2130413045712.FITS → Qatar-2130413084512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-130413.log (789923339683...), FinalLightCurve_Qatar-2 b_2013-04-12.png (3c56c5ceb679...), FinalLightCurve_Qatar-2 b_2013-04-12.csv (7a19c79fb334...)

Compute resources

Wall time 142.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 16:04Z.